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what is the best python parametrized unit test

Parameterized unit testing allows you to run the same test logic with different input values. This helps to avoid duplicating test code and can be very useful for covering a variety of test cases without writing a new test method for each one.

In Python, the `unittest` module provides a base framework for unit testing, but it doesn't have built-in support for parameterized testing. However, there are several ways to achieve parameterized testing in Python.

1. Using `unittest.subTest` :

The `subTest` context manager in the `unittest` framework allows you to execute portions of your test function with different parameters.

python

```
import unittest

class MyTest(unittest.TestCase):
    def test_numbers(self):
        for i, j in [(1, 2), (3, 4), (5, 6)]:
            with self.subTest(i=i, j=j):
                self.assertNotEqual(i, j)
```

 Copy code

2. Using `parameterized` library:

The `parameterized` library is popular and can be used with `unittest` and other testing frameworks like `pytest`.

First, install it:

nginx

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```
pip install parameterized
```

Then, use it in your tests:

```
python
```

```
from parameterized import parameterized
import unittest
```

= ChatGPT 5.2 

```
(1, 2),
(3, 4),
(5, 6),
])
def test_numbers(self, i, j):
    self.assertNotEqual(i, j)
```

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3. Using pytest :

pytest is another popular testing framework in Python that supports parameterized testing out of the box.

First, install it:

```
nginx
```

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```
pip install pytest
```

Then, write parameterized tests like:

```
python
```

 Copy code

```
import pytest

@pytest.mark.parametrize("i,j", [(1, 2), (3, 4), (5, 6)])
def test_numbers(i, j):
    assert i != j
```

Which method is the "best" can depend on your requirements:

- If you're already using `unittest` and don't want to add external dependencies, `subTest` might be a good option.
- If you're okay with adding an external dependency and are using `unittest`, the `parameterized` library provides a more concise way to define parameterized tests.
- If you're open to using a different testing framework, `pytest` is powerful and provides parameterization and many other features.

Given its simplicity, flexibility, and feature set, many developers consider `pytest` to be the best for parameterized unit tests in Python.